Troubleshooting a Document Assistant

Empty or poor answers

If the assistant returns no useful answer, first confirm that the PDFs were parsed. Scanned documents contain images instead of text, so the extractor returns empty strings and nothing is indexed. Such files need optical character recognition before ingestion. Next confirm that the vector collection actually contains records and that the count grows after ingestion. A collection that is empty at query time usually means ingestion wrote to a different database directory than the one the query script reads.

Unrelated passages are retrieved

When retrieval returns passages that have nothing to do with the question, check that the same embedding model is used for ingestion and for queries. Mixing two models produces vectors in different spaces and the nearest neighbours become meaningless. Also inspect the chunk size. Chunks that are too large mix several topics and match many questions weakly, while chunks that are too small lose context. Rephrasing a vague question with the specific terms used in the documents often improves results.

Calibrating the distance threshold

The distance threshold decides when the assistant refuses. To calibrate it, run the test questions with a debug flag that prints the distance of every retrieved chunk. Record the distances of correct matches and the distances returned for questions that are out of scope. Place the threshold between the two groups. Lower the threshold only after reviewing examples, because a strict threshold refuses questions the corpus can answer, while a loose threshold lets unrelated passages reach the model. Re-check the threshold whenever the embedding model or the chunk size changes.